崑山科技大學日間部電機工程系研究所碩士班課程科目表

Kun Shan University Department of Electrical Engineering Graduate School Master Class Curriculum (Day Division)

課程名稱（講授時數－實習時數－學分數）

Course Name (Lecture Hours – Practice Hours – Credits)

113.04.18 系課程委員會通過
113.04.26 院課程委員會通過
113.05.13 校課程委員會通過

113 學年度入學適用

Apply to the freshmen enrolled in 113

第一學年 First Academic Year		第二學年 Second Academic Year	
第一學期 First Semester	第二學期 Second Semester	第一學期 First Semester	第二學期 Second Semester
必修科目(共 5 學分) General Education Required Courses (5Credits)			
1. 校訂必修課程：Compulsory Courses of School			
學術倫理 Academic ethics (1-0-1)			
2. 院訂必修課程：(不列入畢業學分) Compulsory Courses of College (Not included in graduation credits)			
工程技術講座(一) Engineering Technology Lectures (I) (2-0-2)	工程技術講座(二) Engineering Technology Lectures (Ⅱ) (2-0-2)		
3. 專業必修課程：Required subjects for majors			
專題討論（一） Seminar (I) (2-0-0)	專題討論（二） Seminar (Ⅱ) (2-0-0)	專題討論（三） Seminar (Ⅲ) (2-0-0)	專題討論（四） Seminar (Ⅳ) (2-0-0)
專業選修課程(選修學分至少 24 學分，並且選修本系課程之學分數至少 18 學分。) Professional elective courses (at least 24 credits of elective courses, and at least 18 credits of elective courses in this department.)			
高等電力電子 Advanced Power Electronics (3-0-3)	類神經網路應用 Application of Artificial Neural Networks (3-0-3)	高等電力系統分析 Advanced Power System Analysis (3-0-3)	配電系統自動化 Distribution System Automation (3-0-3)
電力品質分析與模擬 Power Quality Analysis and Simulation (3-0-3)	切換式電源轉換器分析與設計 Analysis and Design of Switching Power Converters (3-0-3)	真空技術與產業分析 Vacuum Technology and Its Related Industry (3-0-3)	電力系統穩定度分析 Power System Stability Analysis (3-0-3)
線性系統理論 Linear System Theory (3-0-3)	電力電子模式分析與控制 Modeling and Control of Power Electronics (3-0-3)	光電半導體元件與應用 Optoelectronic Semiconductor Components and Applications (3-0-3)	數位電路設計實務 Digital Circuit Design Practice (3-0-3)
高等半導體元件物理 Advanced Semiconductor Devices Physics (3-0-3)	非線性控制 Nonlinear Control (3-0-3)	伺服電機控制系統 Control system on Servo Motor (3-0-3)	影像處理技術 Image Processing Technology (3-0-3)
馬達模式與控制 Modeling and control of Electric Motor (3-0-3)	半導體技術模擬 Semiconductor Technology Simulations (3-0-3)	模糊控制應用 Fuzzy Control Applications (3-0-3)	光電工程與應用 Application of Photovoltaic Engineering (3-0-3)

第一學年 First Academic Year		第二學年 Second Academic Year	
第一學期 First Semester	第二學期 Second Semester	第一學期 First Semester	第二學期 Second Semester
科技論文寫作 Writing Scientific Papers (3-0-3)	電力暫態分析與模擬 Power Transient Analysis and Simulation (3-0-3)	材料分析 Materials Analysis (3-0-3)	電力系統運轉控制 Power System Operation Control (3-0-3)
電子電路設計實務 Practical design of electronic circuits (3-0-3)	薄膜工程與應用 Thin Film Engineering and Application (3-0-3)	太陽光電工程 Solar Photovoltaic Engineering (3-0-3)	雷射工程 Laser Engineering (3-0-3)
顯示器專論 Monitor Monograph (3-0-3)	太陽電池開發與應用 Development and Application of Photovoltaic Modules (3-0-3)	生物光學 Biopics (3-0-3)	偏振光學特論 Special Treatise on Polarization Optics (3-0-3)
光電化學特論 Special Treatise on Photo Electrochemistry (3-0-3)	LED 照明專論 LED lighting monograph (3-0-3)	控制系統設計與實務 Control System Design & Practice (3-0-3)	半導體工程與應用 Semiconductor Engineering and Applications (3-0-3)
論文 6 學分 Thesis 6 credits			
最低畢業學分：30 學分(不含校訂必修課程及院訂必修課程) Minimum graduation credits: 30 credits (excluding required courses set by the school and required courses set by the college)			

註：

1. 論文課程為必修 6 學分，需通過論文口試。The thesis course is a compulsory 6-credit course and requires passing the thesis oral examination.
2. 最低畢業學分為 30 學分，其中包含論文 6 學分及專業選修課程至少 24 學分。The minimum graduation credits are 30 credits, including 6 credits of thesis and at least 24 credits of professional elective courses.
3. 工程技術講座(一)(二)、學術倫理為必修，但不列入畢業學分數計算。Engineering Technology Lectures (I) (II) and Academic Ethics are compulsory courses, but are not included in the calculation of graduation credits.

4.學術倫理之實施與學分取得依課務組網頁之「學術倫理教育實施要點」辦理。The implementation of academic ethics and the acquisition of credits are handled in accordance with the "Key Points for the Implementation of Academic Ethics Education" on the website of the academic affairs team.